

WEEK 2 – POWER BI SALES PERFORMANCE ANALYSIS.

Superstore Sales Performance and Business Insights Dashboard

INTRODUCTION:

Microsoft Power BI is a business intelligence tool that helps you analyze and visualize data. The dataset was prepared and organized in Power BI. Different visual elements such as cards, charts, slicers, and tables are used to present the data.

OBJECTIVES:

The main objectives of this project are:

- *To load the Superstore Sales Dataset into Power BI.
- *To organize the data for analysis.
- *To create a suitable data model.
- *To develop useful DAX measures.
- *To study sales and profit performance.
- *To analyze category and regional performance.
- *To examine sales changes over time.
- *To create an interactive Power BI dashboard.
- *To obtain useful information from the dataset.

DATASET OVERVIEW:

The Superstore Sales Dataset contains details about customer orders, including the date, time, and location of the order, as well as the product details and the salesperson who handled the order.

Important Fields;

- *Order Date – Date when the customer placed the order.

- *Customer ID – Identifies each customer.
- *Customer Name – Name of the customer.
- *Product ID – Unique product identification.
- *Product Name – Name of the product.
- *Category – Main product category.
- *Sub-Category – Detailed product classification.
- *Sales – Amount generated from sales.
- *Quantity – Number of products purchased.
- *Profit – Profit obtained from sales.
- *Region – Geographical area.
- *Segment – Type of customer segment.

DATA MODELING AND PREPARATION:

The Superstore dataset was imported into Power BI a
A separate Date Table was created to perform time-ba
Relationships were established between the tables so

Power BI Data Model and Relationships;

TIME-BASED ANALYSIS:

A Date Table was created to analyze sales according to

The Date Table contains fields such as:

- *Date.
- *Year.

- *Month Name.

- *Month Number.

The Date Table was connected with the Order Date field.

Date Table in Power BI

DAX CALCULATIONS:

DAX (Data Analysis Expressions) was used to create calculations.

The main measures used in the analysis are:

- *Total Sales.

- *Total Profit.

- *Total Orders.

- *Total Quantity.

- *Average Sales.

- *Profit Margin %.

These calculations help convert the raw dataset into useful information.

DAX Measures in Power BI

REPORT VISUALIZATIONS;

Different Power BI visualizations were selected to present the data.

1.Sales and Profit KPI Cards.

KPI cards were used to display important overall values.

These cards provide a quick summary of business performance.

2.Category Sales Comparison.

A column chart was used to compare sales among different categories.

3.Regional Profit Analysis.

A chart was used to compare profit across different regions.

4.Monthly/Yearly Sales Trend.

A line chart was created to display sales changes over time.

5.Interactive Slicers.

Slicers were added to make the dashboard interactive.

The main filters include:

- *Year
- *Category
- *Region
- *Segment

Users can select different values and view the corresponding data.

6.Decomposition Tree.

A decomposition tree was included to break down Total Sales.

DASHBOARD DESIGN:

The final Power BI dashboard combines multiple visualizations.

The dashboard contains:

- *KPI Cards
- *Sales Category Chart
- *Regional Profit Chart
- *Sales Trend Chart
- *Slicers

*Decomposition Tree

The combination of these components allows users to vi

Final Power BI Dashboard

FINDINGS FROM THE ANALYSIS:

The dashboards from the Superstore dataset.

Sales Performance

Total Sales and Average Sales provide an overview of the

Profitability

Total Profit and Profit Margin help in understanding the

Product Category Analysis

Category-wise sales comparison helps identify differences

Regional Performance

The regional profit chart helps compare profitability be

Time-Based Performance

The sales trend chart helps observe how sales change ac

Customer Segment Analysis

The Segment slicer allows the dashboard to be viewed a

Detailed Sales Analysis

The decomposition tree provides a detailed breakdown

BENEFITS OF THE DASHBOARD

The Power BI dashboard provides several benefits:

***Simple Data Understanding** – Complex data can be unders

***Interactive Reporting** – Users can interact with the dashbo

- ***Performance Monitoring** – Important sales and profit values
- ***Easy Comparison** – Categories and regions can be compared
- ***Trend Identification** – Sales trends can be studied over different periods
- ***Better Data Analysis** – DAX measures provide useful calculations
- ***Quick Insights** – Important business information can be viewed at a glance

CONCLUSION

The Superstore Sales Dataset was successfully analyzed using Power BI. Different visualizations, including KPI cards, charts, slicers, and tables, were created to provide a comprehensive view of the data. This project demonstrates how Power BI can be used to analyze and visualize large datasets effectively.

TOOLS AND TECHNOLOGIES USED

Visualization Tool: Microsoft Power BI.

Data Preparation: Power Query.

Data Modeling: Power BI Relationships and Tables.

Calculation: DAX.

Visualizations: KPI Cards, Column Charts, Line Charts, Slicers, Decomposition Tree.

Dataset: Superstore Sales Dataset.